

The prime-implicants are those terms not checked in the table.
 To convert from decimal notation to binary, convert all decimal numbers in the term to binary and then insert a dash in those positions designated by the numbers in parentheses.

Thus 0, 1 (1) is converted to binary as 0000, 0001; a dash in the first position of either number results in (000-). Similarly, 0, 2, 8, 10 (2, 8) is converted to the binary notation from 0000, 0010, 1000, and 1010, and a dash inserted in position 2 and 8, to result in (-0-0). [Note:] The numbers in the next section below must be greater for the combination to take place.

Example 3-14: Determine the prime-implicants of the function:

$$F(w, x, y, z) = \Sigma(1, 4, 6, 7, 8, 9, 10, 11, 15)$$

Table 3-7 Determination of prime-implicants for Example 3-14

(a)	(b)	(c)
0001 1 ✓	1, 9 (8)	0, 1 (1)
0100 4 ✓	4, 6 (2)	0, 2 (2)
1000 8 ✓	8, 9 (1)	0, 8 (8)
	8, 10 (2) ✓	8, 9, 10, 11 (1, 2)
0110 6 ✓	6, 7 (1)	
1001 9 ✓	9, 11 (2) ✓	
1010 10 ✓	10, 11 (1) ✓	
0111 7 ✓	7, 15 (8)	
1011 11 ✓	11, 15 (4)	
1111 15 ✓		

Decimal	Prime-implicants Binary	Term
1, 9 (8)	- 0 0 1	$\bar{w}\bar{x}z$
4, 6 (2)	0 1 - 0	$w\bar{x}z$
6, 7 (1)	0 1 1 -	$w\bar{x}y$
7, 15 (8)	- 1 1 1	$\bar{w}yz$
11, 15 (4)	1 - 1 1	wyz